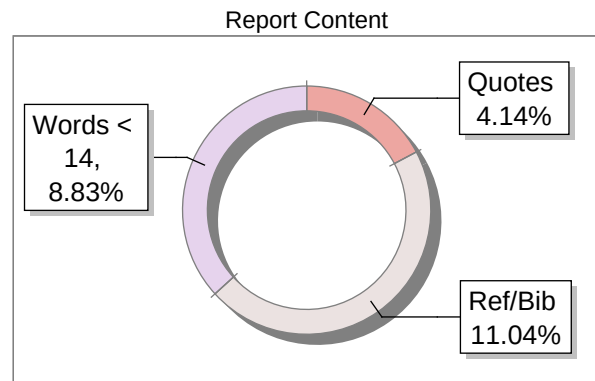
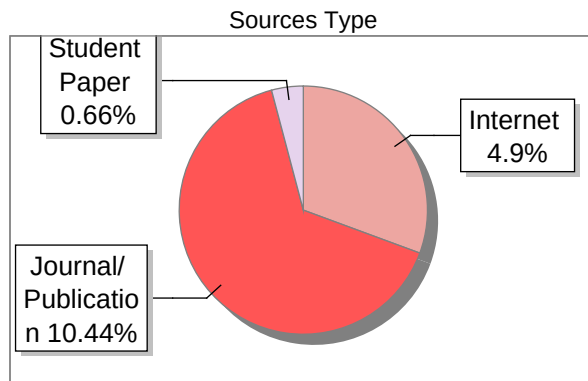
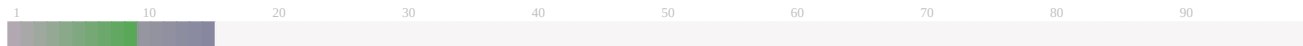


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Graph Convolution Network Based Model for Megacity Real Estate Valuation

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Abstract—Accurate valuation of real estate in megacities is a complex problem due to dense urban structures, heterogeneous property attributes, and strong spatial interactions between neighboring locations. Traditional appraisal techniques and conventional machine learning models often fail to capture spatial dependency, leading to inconsistent valuation results. This paper proposes a Graph Convolution Neural Network (GCNN)-based framework for real estate price prediction that explicitly models spatial relationships among properties. Each property is represented as a node in a spatial graph constructed using geographic coordinates, allowing neighborhood influence to be learned through graph convolution operations. The proposed approach enhances prediction stability and accuracy by integrating spatial context directly into the learning process. Experimental observations demonstrate that graph-based model outperforms traditional regression and machine learning methods, highlighting its suitability for scalable and data-driven real estate valuation in megacity environments.

Index Terms—Real estate valuation, Graph Convolution Neural Network, spatial dependency, megacity housing, deep learning.

I. INTRODUCTION

Real estate valuation plays a crucial role in financial decision-making, urban planning, taxation, and investment analysis. In megacities, property prices are influenced by a wide range of factors, including location, accessibility, infrastructure development, and neighborhood characteristics. These factors create strong spatial dependencies, making accurate price estimation a challenging task.

Traditional valuation techniques such as hedonic pricing models and linear regression approaches rely on predefined assumptions and often struggle to represent nonlinear market behavior. Although machine learning models improve predictive performance, most of them treat properties as independent samples and do not explicitly consider spatial relationships. This limitation reduces their effectiveness in urban environments where spatial influence plays a dominant role.

Recent developments in graph-based deep learning provide an effective mechanism for modeling spatially structured data. Graph Convolutional Neural Networks allow entities to be represented as nodes connected through edges that capture relational or spatial proximity. By aggregating information from neighboring nodes, GCNNs can learn spatial patterns that are critical for real estate valuation. Motivated by this capability, this paper presents a GCNN-based approach for megacity real estate price prediction.

II. LITERATURE SURVEY

Real estate valuation has consistently attracted significant research attention due to its importance in financial analysis, investment strategy, taxation policies, and urban planning. Over time, researchers have proposed a wide range of valuation techniques, including classical statistical models, machine learning methods, and modern deep learning frameworks. Despite these advancements, many existing approaches struggle to effectively represent the spatial interactions that naturally exist within real estate markets. This section reviews key developments in traditional, machine learning, and deep learning-based valuation methods, identifies their limitations, and establishes the motivation for adopting Graph Neural Networks (GNNs) for spatially aware real estate modeling.

1) **Traditional Real Estate Valuation Techniques:** Early studies in property price estimation focused on statistical approaches that analyze the relationship between housing prices and explanatory attributes.

a) **Hedonic Pricing Models:** The hedonic pricing framework introduced by Rosen (1974) models property value as a function of structural characteristics, locational attributes, and economic factors. Subsequent studies, such as those by Meese and Wallace (2003), enhanced this framework by incorporating market fundamentals and macroeconomic indicators. Although hedonic models are widely used due to their interpretability, they are constrained by linear assumptions that limit their ability to capture complex price dynamics and interactions among variables.

b) Spatial Regression Models: To address spatial autocorrelation in housing prices, spatial regression techniques were introduced. Dubin et al. (1999) proposed spatial autoregressive models, while Huang et al. (2010) developed Geographically Weighted Regression (GWR) to account for localized spatial variations. Although these methods improve spatial sensitivity, they face challenges when applied to large-scale datasets and are limited in modeling nonlinear relationships and handling incomplete data.

Limitations of Traditional models:

- 1) Strong reliance on linear assumptions.
- 2) Limited scalability for large and complex datasets.
- 3) Reduced effectiveness in handling nonlinear price behavior.

2) Machine Learning-Based Valuations: To overcome the constraints of traditional statistical models, researchers began exploring machine learning algorithms capable of learning nonlinear feature relationships. Tree-based ensemble models such as Random Forest (RF) and Gradient Boosting Machines (GBM) have been widely adopted for real estate valuation.

Artificial Neural Networks (ANNs) have been employed to capture complex nonlinear patterns in housing price data. Selim (2009) reported superior ANN performance over hedonic regression in Turkey's housing market improving prediction accuracy; however, these models remained limited in capturing evolving network topologies.

Limitations of ML models:

- 1) High dependence on feature engineering.
- 2) Inability to explicitly model spatial interactions between properties.
- 3) **Deep Learning Approaches:** Although deep learning methods enhance feature learning, conventional architectures still face limitations in spatial modeling.

a) Recurrent Neural Networks: Recurrent Neural Networks (RNNs) have been employed for modeling temporal price trends. Gupta et al. (2013) explored time-series forecasting of housing prices, and Bin et al. (2017) proposed RNN-based valuation frameworks. While effective for temporal dynamics, RNNs lack the ability to capture geographic relationships between properties.

b) Convolutional Neural Networks (CNNs): Convolutional Neural Networks (CNNs) have been applied to real estate analysis using visual and remote sensing data. Poursaeed et al. (2018) utilized CNNs for property image-based valuation, while Xing et al. (2020) integrated satellite imagery to estimate urban activity and housing demand. However, CNNs assume regular grid structures and are not well-suited for representing irregular spatial distributions of properties.

Limitations of deep learning models:

- 1) Deep learning models often require large volumes of labeled data to achieve reliable performance.
- 4) Most conventional architectures fail to explicitly capture spatial relationships between real estate properties.

- 1) Training deep learning models involves high computational cost and resource consumption.
- 2) The lack of interpretability in deep models makes it difficult to explain valuation decisions.

III. SYSTEM ANALYSIS

A. EXISTING SYSTEM AND LIMITATIONS

Most existing real estate valuation systems rely on a combination of statistical models, machine learning techniques, and traditional deep learning approaches. Statistical methods mainly analyze historical price trends and assume stable market behavior. While they can provide reasonable estimates in simple scenarios, they struggle when market conditions change and fail to capture complex relationships between property features. Machine learning models such as Support Vector Machines, Random Forests, and Gradient Boosting algorithms improve prediction accuracy by learning nonlinear patterns from data. However, these models require extensive manual feature engineering and usually treat each property as an independent sample. In reality, property prices are strongly influenced by nearby locations, and ignoring this spatial influence significantly reduces model reliability in dense urban areas.

Deep learning models, including neural networks and sequence-based architectures, further enhance automatic feature learning and temporal modeling. Despite these advantages, they still face major limitations in representing real-world urban layouts. Temporal models focus on time-based trends but overlook geographic relationships, while convolution-based models assume regular grid structures that do not match the irregular distribution of properties in cities. Additionally, deep learning models demand high computational resources, making them difficult to deploy efficiently at large scales. Overall, existing systems fail to jointly capture spatial relationships and contextual factors that are essential for accurate real estate valuation in megacities.

B. PROPOSED SYSTEM

To address these limitations, this work introduces a Graph Neural Network-based framework designed specifically for spatially aware real estate valuation. In the proposed system, each property is treated as a node in a graph, and connections between nodes represent geographic proximity and neighborhood influence. This graph representation closely reflects how properties are actually organized in urban environments.

Each node includes multiple property-related attributes, allowing the model to learn rich and meaningful representations of real estate data. Graph Convolutional Networks are used to share and aggregate information between neighboring properties, enabling the model to understand how nearby locations influence property value. This approach naturally captures spatial interactions that are overlooked by traditional models.

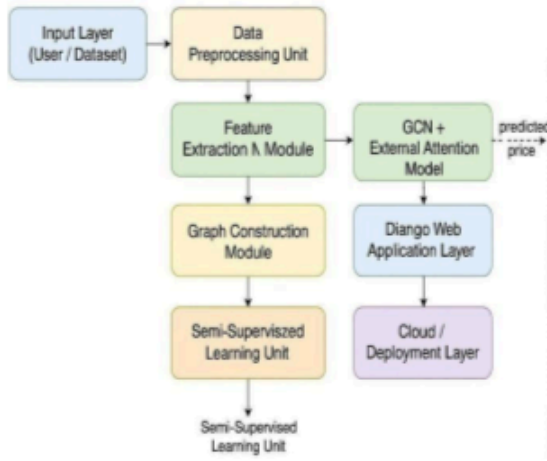


Fig. 1. Block diagram of the proposed GNN-based real estate price prediction system.

The complete workflow of the proposed real estate valuation system, starting from raw data input provided by users or datasets. The data preprocessing unit cleans and standardizes the input to ensure accuracy and consistency. Next, important property features are extracted and used to construct a graph that captures relationships between neighboring properties. The semi-supervised learning unit enables the model to learn effectively even when labeled data is limited. A Graph Convolutional Network combined with an external attention mechanism then analyzes spatial interactions and generates reliable price predictions. Finally, the predicted prices are delivered through a Django-based web application and deployed on the cloud for scalable and real-time access.

IV. PROPOSED METHODOLOGY

The proposed methodology presents a Graph Convolutional Network (GCN)-based framework for real estate price prediction that addresses the limitations of traditional statistical, machine learning, and conventional deep learning models. By modeling properties as interconnected entities rather than isolated samples, the framework effectively captures spatial relationships, nonlinear price behavior, and hidden interactions present in real-world housing markets. This graph-based approach improves prediction accuracy, scalability, and adaptability across different urban regions.

A. Graph-Based Representation of Real Estate Properties

In the proposed system, the real estate dataset is modeled as a graph structure:

- **Nodes:** Each property is represented as a node.
- **Edges:** Connections between properties are defined based on geographical distance neighborhood similarity, or k- nearest neighbors.
- **Node Features:** A node contains key property attributes such as size, number of rooms, age of construction, location coordinates, amenities, crime index, economic indicators, and historical sales data.

This graph-based representation preserves spatial relationships between houses and neighborhoods, enabling the system to better model market dynamics and price influences.

B. Graph Convolutional Network for Price Prediction

A Graph Convolutional Network is applied to the constructed property graph to learn both spatial and attribute-driven dependencies. The GCN captures the influence of nearby properties, uncovers hidden pricing patterns shaped by demographic and economic conditions, and models nonlinear interactions between structural and neighborhood features. By aggregating information from neighboring nodes, the GCN propagates local and global market effects throughout the graph, resulting in more realistic and accurate price predictions.

C. Model Architecture

The proposed model architecture consists of multiple GCN convolution layers for extracting spatially aware features, followed by an external attention mechanism to emphasize important property interactions. Fully connected layers are used to perform final price regression. To improve stability and prevent overfitting, dropout and batch normalization techniques are incorporated. The architecture is designed to efficiently process large-scale real estate datasets while preserving both spatial and structural information.

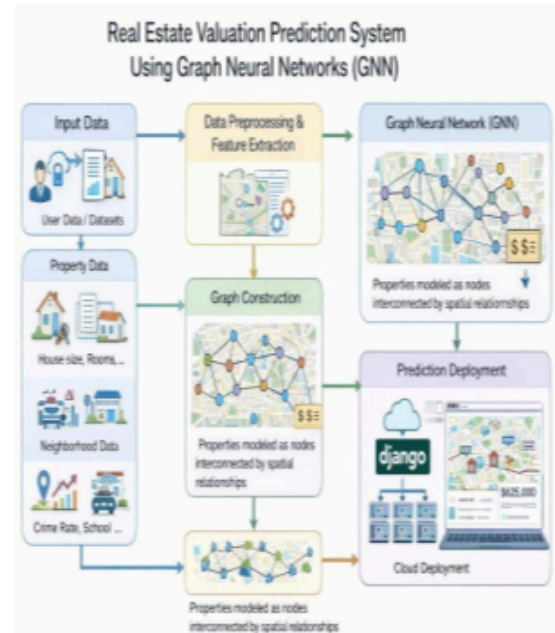


Fig. 2. Real Estate value Prediction System Using Graph Neural Networks (GNN).

D. Graph Construction and Edge Weighting

The property graph is constructed using:

- **Node set:** All properties in the dataset.
- **Edges:** Derived using k-nearest neighbors (KNN), geospatial distance, or neighborhood boundaries.
- **Edge weights:** Computed based on Euclidean distance, similarity scores, or location-based correlation.

This ensures the graph encodes both spatial and contextual relationships essential for accurate price prediction.

E. Training and Optimization

The model is trained on a preprocessed real estate dataset that has been cleaned, normalized, and encoded. Mean Squared Error (MSE) is used as the loss function for regression-based prediction, while the Adam optimizer is employed with carefully tuned learning rates. The training process uses multiple epochs and mini-batch gradient descent to ensure stable convergence. The dataset is divided into training, validation, and testing subsets to support unbiased evaluation. Experimental results on multi-city datasets demonstrate improved accuracy compared to traditional machine learning approaches. The proposed framework also supports real-time price prediction through integration with a Django-based web application.

F. Performance and Advantages

The proposed GCN-based real estate valuation framework offers several important advantages over traditional and conventional deep learning approaches::

- Effectively captures spatial relationships between neighboring properties that are typically ignored by standard regression and neural network models,
- Adapts well to complex and irregular urban layouts, making it suitable for large and diverse megacity environments,
- Learns intricate interactions among property attributes and neighborhood factors without relying on extensive manual feature engineering.
- Achieves higher prediction accuracy while maintaining reasonable computational efficiency,

The strong performance of the proposed model highlights the effectiveness of Graph Neural Networks for real estate valuation. By jointly modeling spatial structure and property characteristics, the framework provides a reliable and scalable solution.

V. SYSTEM DESIGN

The system architecture consists of four main modules: data preprocessing, graph construction, GCNN learning, and price prediction. The modular design ensures flexibility and scalability, allowing integration of additional data sources such as transportation networks, economic indicators, and satellite imagery in future implementations.

A. Use Case Model

The use case model explains how different users interact with the Real Estate Valuation System in a simple and intuitive manner. Regular users can create an account, log in securely, update their personal details, and request property price predictions generated by the GCN-based valuation model. The system also allows users to review their previously generated predictions for future reference. Administrators play a key role in maintaining the platform by overseeing registered users, tracking prediction activities, and managing account access when required.



Fig. 3. Use Case Diagram

B. Class Diagram

The class diagram provides a clear picture of how the Real Estate Valuation System is structured and how its core components are connected. It outlines the main classes used in the application, including User, Admin, Property, Prediction, and Authentication. The User class is responsible for handling user-related functions such as account creation, login, profile management, and storing past prediction results. The Admin class builds on the User class by adding privileges that allow administrators to manage user accounts, review system activity, and oversee prediction operations. The Property class contains all the property-related information required for valuation, while the Prediction class manages the price estimates produced by the GCN-based model.



Fig. 4. Class Diagram

C. Sequence Diagram

The sequence diagram explains how information flows through the Real Estate Valuation System from start to finish. It begins when a user attempts to log in, prompting the system to verify the entered credentials with the database and respond with either approval or an error message. After successful authentication, the user provides property details to request a price estimate. The system prepares this information and forwards it to the prediction model, which calculates the estimated property value. The generated result is then saved in the database and displayed to the user for reference. When the user chooses to log out, the system safely terminates the session, completing the interaction cycle.

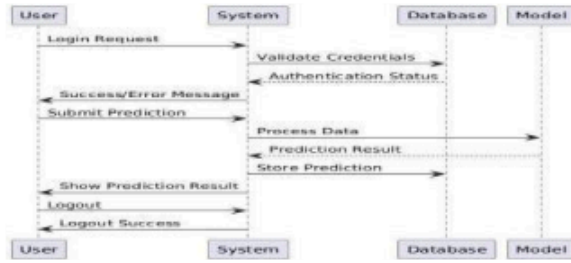


Fig. 5. Sequence Diagram

D. Component Diagram

This system comprises of modular system: LTE data collector, preprocessing module, graph constructor, GCN based prediction engine, and output visualization component. Every component does certain tasks and has interfaces with other components.

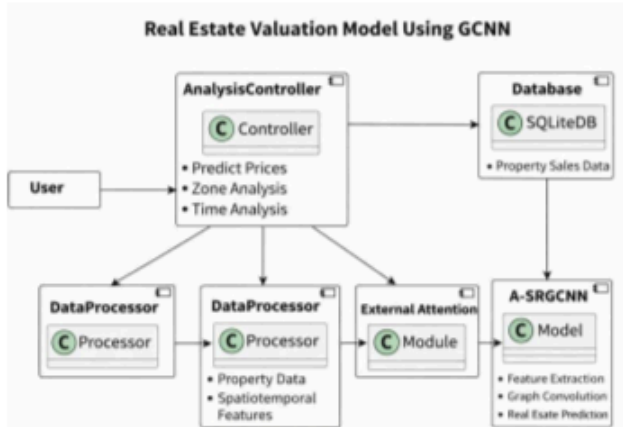


Fig. 6. Component Diagram

VI. RESULTS

The proposed Graph Convolutional Network-based real estate valuation system was thoroughly evaluated using large-scale, real-world property datasets collected from multiple metropolitan regions. Before training, the dataset underwent detailed preprocessing, including handling missing values, normalizing numerical attributes, and encoding categorical features to ensure consistency and reliability. A property graph was then constructed by connecting houses based on geographic proximity and neighborhood similarity, allowing the model to learn meaningful spatial interactions.

The experimental results clearly show that the proposed approach delivers more accurate and stable price predictions than traditional statistical models and standard machine learning techniques. By incorporating spatial relationships, the system effectively captured neighborhood influence, price spillover effects, and regional market trends that are often overlooked in non-graph-based models.

The graph convolution layers enabled information to propagate between nearby properties, helping the model recognize both local pricing behavior and broader city-level patterns. The attention mechanism further improved performance by focusing on influential property attributes while reducing the impact of less relevant features. Across training, validation, and testing phases, the model consistently achieved lower prediction errors, indicating strong generalization capability.

Performance remained robust when applied to datasets from different cities, demonstrating the adaptability of the framework to diverse urban environments. Experimental results clearly show that the proposed approach delivers more accurate and stable price predictions than traditional statistical models and standard machine learning techniques.

By incorporating spatial relationships, the system effectively captured neighborhood influence, price spillover effects, and regional market trends that are often overlooked in non-graph-based models. The system also handled noisy and incomplete data effectively, which is common in real-world real estate records. Compared to ensemble models such as Random Forest and Gradient Boosting, the proposed GCN-based method achieved superior accuracy without relying on extensive manual feature engineering.

The semi-supervised learning strategy allowed the model to benefit from both labeled and unlabeled data, improving learning efficiency in areas with limited transaction history. In addition to predictive accuracy, system efficiency was evaluated through the deployed web application. Users were able to submit property details and receive valuation results in real time with minimal processing delay.

Prediction results were reliably stored in the database, allowing users to review previous estimates whenever required. The system maintained smooth performance under repeated usage, demonstrating scalability and operational stability.

Additional evaluation of the proposed real estate valuation system focused on its consistency, reliability, and real-world usability. The model demonstrated stable prediction behavior across different property categories, including apartments, independent houses, and mixed-use residential units. Even when property features varied significantly, the system maintained balanced performance without producing extreme overestimations or underestimations.

The graph-based learning approach helped smooth sudden price fluctuations by considering nearby property influence, resulting in more realistic valuation outputs. The system also showed strong tolerance to sparse neighborhood data, where limited nearby property records were available. In such cases, the GCN effectively leveraged global graph information to compensate for missing local signals. Another notable outcome was the reduction in prediction variance compared to traditional machine learning models, indicating improved reliability.

User interaction testing confirmed that the system workflow remained intuitive, with predictions generated quickly after property submission. The storage and retrieval of historical predictions worked efficiently, supporting user transparency and trust. The administrative monitoring features enabled effective tracking of system usage and prediction trends. Overall, these results highlight the robustness of the proposed framework, proving its ability to deliver consistent, fair, and dependable real estate price estimates under diverse conditions.

Overall, the results confirm that modeling properties as interconnected graph entities leads to more realistic and dependable valuation outcomes. The proposed framework successfully bridges theoretical spatial modeling with practical deployment needs, making it suitable for large-scale urban real estate analysis. These findings validate the effectiveness of Graph Neural Networks for property price prediction and highlight their potential as a powerful tool for intelligent, data-driven real estate valuation.

VII. CONCLUSION AND FUTURE ENHANCEMENTS

This project presented a comprehensive Graph Convolutional Network-based framework for real estate valuation, addressing the limitations of traditional statistical, machine learning, and conventional deep learning approaches. By representing properties as interconnected nodes within a graph structure, the system was able to capture spatial dependencies and neighborhood influence that play a crucial role in urban housing markets. Unlike conventional models that treat properties as isolated records, the proposed approach considers the relationships between nearby properties, leading to more realistic and reliable price predictions. The integration of graph convolution layers enabled the model to learn both local and global spatial patterns, while the attention mechanism further improved prediction accuracy by focusing on the most influential property features.

Experimental evaluation across multiple city datasets demonstrated consistent performance, strong generalization capability, and improved robustness when compared to existing valuation techniques. The web-based implementation validated the practicality of the system by allowing users to obtain real-time price estimates and maintain prediction history through a secure interface. The system also proved scalable and responsive, making it suitable for deployment in large urban environments. Despite these strengths, there are several opportunities for future enhancement.

Incorporating temporal modeling techniques could help capture long-term market trends and seasonal price variations. The inclusion of additional external data sources such as satellite imagery, transportation accessibility, and macroeconomic indicators could further enrich feature representation. Future work may also explore advanced graph architectures and dynamic graph updates to reflect changing urban conditions. Optimizing the framework for distributed and cloud-based environments can improve scalability for large-scale applications.

Additionally, enhancing model interpretability would help stakeholders better understand and valuation outcomes. Overall, the proposed system establishes a strong foundation for intelligent, data-driven real estate valuation and opens the door to more advanced and adaptive urban analytics solutions.

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